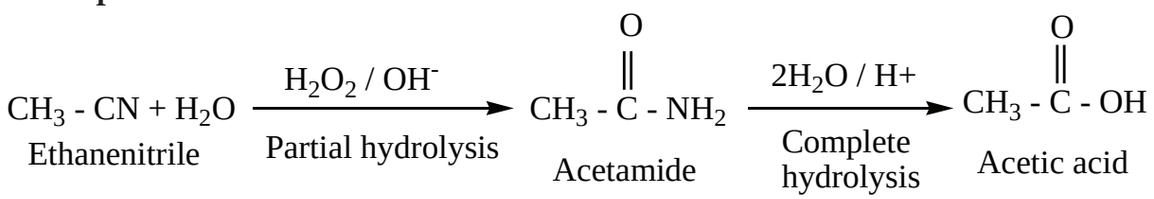
13.4.4 Chemical properties

1. Hydrolysis

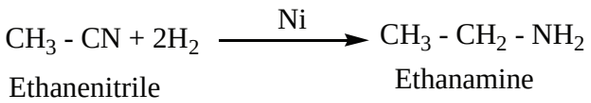
On boiling with alkali (or) a dilute mineral acid, the cyanides are hydrolysed to give carboxylic acids.

For example



2. Reduction

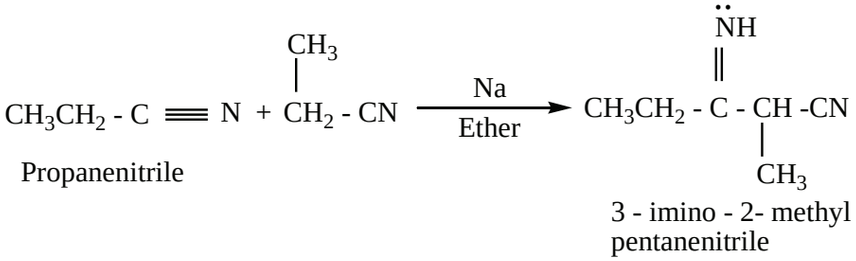
On reduction with $LiAlH_4$ (or) Ni / H_2, alkyl cyanides yields primary amines.



3. Condensation reaction

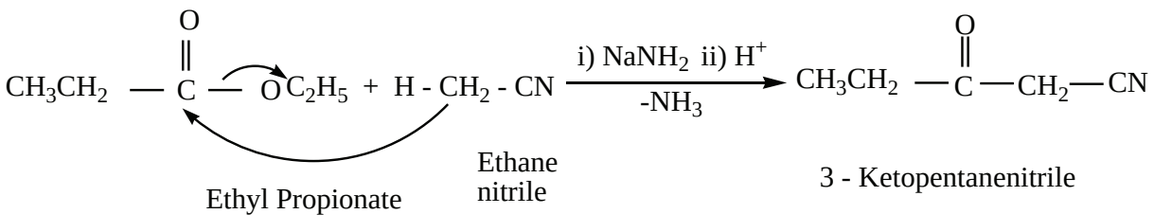
a) Thorpe nitrile condensation

Self condensation of two molecules of alkyl nitrile (containing α–H atom) in the presence of sodium to form iminonitrile.



b) The nitriles containing α-**hydrogen** also undergo condensation with esters in the presence of sodamide in ether to form ketonitriles. This reaction is known as " Levine and Hauser" acetylation

This reaction involves replacement of ethoxy (OC_2H_5) group by methylnitrile $(-CH_2CN)$ group and is called as cyanomethylation reaction.



XII U13-Organic Nitrogen Compounds.indd 225 11-01-2021 16:30:43